Appendix 3 Papua New Guinea—Debt Sustainability Analysis[1]

Papua New Guinea (PNG) is assessed using the low-income country debt sustainability analysis (LIC DSA).[2] While PNG remains at a moderate risk of debt distress, public debt sustainability has improved compared to the 2010 DSA assessment because of a stronger growth outlook. Under the baseline scenario, all external debt sustainability indicators remain well below their applicable thresholds. Stress tests indicate that the threshold for the present value of external public debt to GDP is breached only if prices of commodity exports in 2011–2013 are substantially below WEO forecasts and the government borrows to finance domestic consumption. Overall public debt should continue its downward path under the baseline. However, contingent liabilities related to the LNG project, arrears to public pension funds, and other domestic liabilities would delay this long-term adjustment. Substantial deterioration in the primary balance would harm PNG's debt performance.

I. Background

PNG's public and publicly guaranteed (PPG) debt levels have fallen dramatically over the past decade. Total public debt declined from 70 to 26 percent of GDP from 2001 to end-2010 (Appendix Table 1). External public debt fell from 50 to 11 percent of GDP during this period (Appendix Table 3a). Multilateral lenders such as the World Bank and the AsDB account for around 75 percent of PPG external debt, while Japan and Australia are the largest of the bilateral creditors. Private sector external debt at end-2010 is estimated at 13 percent of GDP.

II. Economic Outlook and Underlying DSA Assumptions

PNG weathered the global financial crisis well due to a large fiscal stimulus, strong domestic demand due to LNG project construction activities, and rebounding commodity prices. Real GDP growth was 7 percent in 2010 after a modest slowdown in 2009, and is projected to accelerate in 2011. PNG's terms of trade and export earnings recovered dramatically from the lows of 2009, and both are projected to improve strongly in 2011 under the baseline scenario. Foreign reserves increased to around US$3.2 billion by March 2011, reflecting large and continuing financial inflows associated with the injection of equity and drawdown of loan funds for the financing of the LNG project by its partners. As indicated in the previous DSA,[3] PNG is vulnerable to external demand and growth shocks. Due to rising commodity prices and strong growth, unfavorable public debt dynamics did not materialize in 2010.

Under the baseline scenario, commodity prices will remain high over the medium term while real GDP will move in line with the LNG sector. During the construction phase of the LNG project, elevated levels of domestic demand and the current account deficit will continue. With the start of LNG production in 2015, real GDP is projected to jump by 20 percent and the current account will move into a large and persistent surplus. Appendix Box 1 summarizes the medium-term macroeconomic framework underlying the DSA.

III. External Debt Sustainability Analysis

Over the projection period, all baseline PPG external debt and debt service indicators stay well below the policy-dependent debt burden thresholds (Appendix Table 3b and Appendix Figure 1). The PV of PPG external debt as a percentage of GDP is on a declining path and stays under 10 percent in the medium term, far below the 30 percent threshold. The PVs of PPG external debt-to-exports and debt-to-revenue ratios are also expected to stay below their applicable thresholds. Since additional financing needs for the LNG project and the current account deficit would be met with projected large FDI inflows, there would be no extra debt burden

[1] Since Papua New Guinea is an IBRD/IDA blend country, this DSA is prepared by Fund staff in consultation with the World Bank and the Asian Development Bank (AsDB) under the IMF-WB DSA framework for Low-Income Countries. The fiscal year of Papua New Guinea is the calendar year.

[2] Papua New Guinea is rated by the World Bank as a weak performer for its policies and institutions for the purposes of the IMF-WB low-income country DSA framework. Hence the applicable thresholds for this category for external public debt are: 30 percent for the present value (PV) of debt-to-GDP ratio; 100 percent for the PV of debt-to-exports ratio, 200 percent for the PV of debt-to-revenue ratio, 15 percent for the debt service-to-exports ratio, and 25 percent for the debt service-to-revenue ratio.

[3] See Papua New Guinea: 2010 Article IV Consultation, IMF Staff Report 10/164.

created. After 2015, the external debt burden is expected to fall even faster due to the projected large increase in exports and revenue from LNG production.

Policy-Based PPG External Debt Burden Thresholds for PNG				
	Thresholds	2010	PNG's ratios 2011-16	2017-31
PV of debt in percent of:				
GDP	30	11.1	8.5	5.4
Exports	100	17.5	12.7	11.6
Revenues	200	41.2	39.6	34.5
Debt service in percent of:				
Exports	15	1.8	1.2	0.9
Revenues	25	4.1	3.7	2.6

External debt sustainability is maintained under an array of possible stress tests, but would be threatened by poor fiscal policy in response to a large terms of trade shock (Appendix Table 3b). Consistent with WEO commodity price forecasts, the baseline scenario predicts PNG's terms of trade to improve in 2011 before settling at a permanently high level. This, combined with the boost from LNG production, makes PNG resilient to a wide range of GDP growth and depreciation shocks. If we instead use the PNG authorities' more conservative estimates for export prices over 2011–2013,[4] and allow the government to borrow to finance the baseline levels of real consumption and imports (contrary to their stated fiscal strategy), then the PV of PPG external debt breaches the threshold of 30 percent of GDP. The price of gold is the most important price for external sustainability before LNG production kicks in.

Delays to the LNG project could generate adverse debt dynamics, even if PPG external debt indicators remain within the thresholds. While it is unrealistic for the LNG project to halt completely, delays to the project would slow down GDP growth and worsen PNG's debt ratios. The historical scenario charts in Appendix Figure 1 can be interpreted as representing PNG's debt dynamics in the absence of LNG construction and production, and without its associated liabilities, assuming that historical levels of real GDP growth are maintained.

IV. Public Debt Sustainability Analysis[5]

In the baseline scenario, PNG's public debt ratio will continue its downward path over the medium and long term (Appendix Table 1 and Appendix Figure 2). This trend is aided by high commodity prices, rapid real GDP growth and the implementation of the authorities' Medium Term Debt Strategy. The public debt-to-GDP ratio is forecast to decrease over the projection period as long as windfall revenues are saved during years of booming commodity prices. Provided these policies are in place, the PV of public sector debt will decline from 27 percent of GDP in 2010 to 13 percent of GDP in 2016 and to 7 percent of GDP by 2031.

Economic growth and saving of commodity tax revenues are the key factors for the projected public debt path as shown in alternative scenarios and bound tests (Appendix Table 2). The public debt-to-GDP ratio would rise to 30 percent in 2031 if there is a negative shock to real GDP growth in 2012–13 with permanently lower revenues and unchanged expenditures until 2031.[6] Keeping the primary fiscal surplus at 2.3 percent from 2011 through 2031 instead of moving into deficits over the medium term as in the baseline would result in the PV of the public debt-to-GDP ratio falling to zero by 2016. This scenario emphasizes the

[4] The stress test uses authorities' prices for gold, copper and oil only. This amounts to an average negative shock to export values of 22 percent relative to the baseline for each of the three years.

[5] Public debt includes domestic central government, external public, and publicly guaranteed debt.

[6] This is a temporary shock to GDP growth, so that the growth rates during 2012–13 are set to the historical average minus one standard deviation. Another permanent GDP growth shock, which could bring the PV of the debt-to-GDP ratio to 31 percent in 2031, is defined as the baseline GDP growth rate minus one standard deviation divided by the square root of the length of the projection period.

sensitivity of the public debt outlook to saving of windfall commodity revenues. Substantial deterioration of the primary balance in future budgets relative to the baseline would negatively affect debt sustainability.

Realization of contingent liabilities related to the LNG project, arrears to public pension funds, and other domestic liabilities would delay the downward trend in public debt ratios. The government's unfunded superannuation liabilities, estimated to be 7 percent of GDP by end-2010, continue to grow, especially given the decision in the 2009 and 2010 budgets not to fully fund the central government's 8.4 percent employer contributions. The government should develop a firm payment schedule for these unfunded liabilities. Contingent liabilities related to the LNG project are estimated to be about 7 percent of GDP in 2015.[7] Although information is limited, attention should be given to the government's possible need to backstop SOE borrowing.[8] Should these factors materialize, unfavorable debt dynamics would not undermine public debt sustainability, but would substantially delay fiscal consolidation (Appendix Table 2 and Appendix Figure 2).[9]

V. Conclusion

PNG remains at moderate risk of debt distress, but PPG external debt sustainability has substantially improved as a result of high commodity prices and upcoming LNG production. The baseline scenario indicates that all external PPG debt burden indicators stay well below their indicative thresholds. External debt sustainability remains resilient to most stress tests, but would be threatened by a large negative terms of trade shock coupled with a poor fiscal response.

Public debt sustainability is sensitive to growth shocks and saving of windfall revenues over the longer term to pay down the debt. The government needs to carefully consider the risk of accumulating public liabilities until LNG taxation revenues are realized. Despite recent changes to the Medium Term Fiscal Strategy to allow 100 percent of windfall revenues to be used for public investment, it would be advisable for the authorities to use some portion of such gains to pay down high interest rate public debt. This would insulate the economy from adverse debt dynamics after possible negative growth shocks. One-off realizations of contingent liabilities would delay the paydown of government public debt, but would not generate unsustainable debt dynamics.

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[7] These liabilities arise from the issue of exchangeable bonds by the PNG government to finance its equity participation (in 2009) and its completion guarantee for debt financing (by 2015). Under the scheme, the creditor acquired exchangeable bonds, amounting to US$1.1 billion, with the option to exchange the bonds for equities in the Oil Search Company. Under the completion guarantee, if the project fails, the government may have to pay up to US$2 billion to the creditor.

[8] From 2008 Article IV DSA estimates based on end-2007 financial statements of nine SOEs, contingent liabilities related to SOEs amount to about 2.5 percent of GDP. Off-balance sheet liabilities, which also may be significant, are not considered. Improving the financial data of SOEs, especially related to contingent liabilities, would help authorities get a better picture of public debt. Staff is supportive of government intentions to move toward a whole-of-government debt stock approach.

[9] The alternative scenario (All Other Liabilities: Pension, SOEs, and LNG contingent liabilities) assumes that pension and SOEs liabilities are to be covered in the 2012 budget (total about 8 percent of 2012 GDP) and all LNG contingent liabilities will materialize in 2015 (about 7 percent of GDP).

Appendix Box 1 Macroeconomic Assumptions Underlying the DSA

- **Real GDP growth** is projected to be 8 percent on average over the medium term, above the historical average of 4 percent, and slow gradually to 3–4 percent in the long run. In the medium term growth in the LNG and nonmineral sectors will offset the decline in copper and petroleum production.
- Construction of the **LNG project** will be finished and production and exports are expected to start in 2015, with a maximum capacity of 6.6 million tons of LNG produced per annum. LNG production is estimated to raise the level of real GDP by about 20 percent in 2015. However, accounting for substantial income outflows, the LNG project is expected to increase annual GNI by about 8 percent.
- **Inflation** is projected to rise to 9 percent in the near term due to increasing domestic demand and near-term capacity constraints, while stabilizing at around 4 percent in the long run.
- The **current account** (including grants) is estimated to be in deficit in 2010 and the deficit is expected to remain sizeable until 2015, primarily reflecting the strong import growth during the construction phase of the LNG project. FDI inflows from foreign shareholders are projected to increase substantially and finance most of the imports.
- The **grant element of loans** is expected to decline. As per capita income rises, the share of external financing provided on concessional terms is expected to decline over the projection period.[1]
- The **primary fiscal balance** is projected to be in surplus of 2.3 percent of GDP in 2011 accounting for spending from the trust accounts. In the medium term primary deficits are anticipated, but these turn into surpluses over the long run as LNG tax revenues are realized.

[1] Grant-equivalent financing (in percent of GDP) decreases over time. From 2016 onwards, the majority of new AsDB disbursements are through OCR facilities, which carry a higher interest rate than ADF loans.

Appendix Table 1 Papua New Guinea: Public Sector Debt Sustainability Framework, Baseline Scenario, 2008–2031

(In percent of GDP, unless otherwise indicated)

	Actual							Estimate	Projections								
	2008	2009	2010	Average	5/	Standard Deviation	5/	2011	2012	2013	2014	2015	2016	2011-16 Average	2021	2031	2017-31 Average
Public sector debt 1/	31.7	32.1	26.5					21.9	18.3	18.3	19.8	14.5	13.8		10.9	8.1	
o/w foreign-currency denominated	13.2	12.9	10.7					9.7	9.4	9.7	9.7	7.2	6.9		6.4	5.6	
Change in public sector debt	-2.3	0.3	-5.6					-4.6	-3.6	0.0	1.5	-5.3	-0.7		-0.3	-0.3	
Identified debt-creating flows	-10.5	8.9	-4.0					-5.0	-2.7	-0.6	1.9	-1.4	0.1		-2.3	-0.6	
Primary deficit	-6.9	7.5	-1.2	-3.4		5.2		-2.3	-1.6	-0.6	2.4	2.9	0.5	0.2	-2.2	-0.5	-0.9
Revenue and grants	32.3	27.5	32.5					33.4	31.4	29.5	25.2	17.6	16.9		18.5	16.5	
of which: grants	5.1	4.0	5.5					5.0	4.4	4.1	3.9	2.7	2.4		1.8	1.1	
Primary (noninterest) expenditure	25.4	35.0	31.3					31.1	29.8	28.9	27.7	20.5	17.4		16.3	16.0	
Automatic debt dynamics	-3.6	1.4	-2.8					-2.7	-1.1	0.0	-0.5	-4.3	-0.4		0.0	-0.1	
Contribution from interest rate/growth differential	-1.7	0.7	-2.0					-2.7	-1.1	0.0	-0.5	-4.3	-0.4		0.0	-0.1	
of which: contribution from average real interest rate	0.4	2.3	0.1					-0.5	0.0	0.3	0.2	-1.0	0.3		0.3	0.2	
of which: contribution from real GDP growth	-2.1	-1.6	-2.1					-2.2	-1.1	-0.4	-0.7	-3.3	-0.7		-0.3	-0.2	
Contribution from real exchange rate depreciation	-1.9	0.7	-0.8					0.0	0.0	0.0	0.0	0.0	0.0		...	...	
Other identified debt-creating flows	0.0	0.0	0.0					0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0					0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0					0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0					0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0					0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	8.2	-8.5	-1.6					0.4	-0.9	0.6	-0.4	-3.9	-0.9		2.0	0.3	
Other Sustainability Indicators																	
PV of public sector debt	...	...	26.9					22.1	18.3	18.0	19.3	14.0	13.2		10.1	7.3	
o/w foreign-currency denominated	...	...	11.1					9.8	9.3	9.4	9.2	6.7	6.4		5.7	4.8	
o/w external	...	...	11.1					9.8	9.3	9.4	9.2	6.7	6.4		5.7	4.8	
PV of contingent liabilities (not included in public sector debt)	...	...	...					...	...	...	...	...	...		...	...	
Gross financing need 2/	1.3	17.7	8.1					4.3	3.2	2.2	5.3	6.5	4.5		0.6	1.3	
PV of public sector debt-to-revenue and grants ratio (in percent)	…	…	82.8					66.2	58.1	61.1	76.5	79.4	78.1		54.8	44.4	
PV of public sector debt-to-revenue ratio (in percent)	…	…	99.5					77.9	67.7	71.0	90.6	93.8	91.0		60.5	47.5	
o/w external 3/	…	…	41.2					34.7	34.4	36.9	43.0	44.8	43.9		33.9	30.9	
Debt service-to-revenue and grants ratio (in percent) 4/	15.3	14.8	9.8					8.2	8.1	7.6	8.2	9.4	10.0		5.8	5.3	
Debt service-to-revenue ratio (in percent) 4/	18.2	17.3	11.7					9.7	9.4	8.8	9.8	11.1	11.7		6.4	5.7	
Primary deficit that stabilizes the debt-to-GDP ratio	-4.6	7.2	4.4					2.3	2.0	-0.6	0.9	8.2	1.2		-2.0	-0.3	
Key macroeconomic and fiscal assumptions																	
Real GDP growth (in percent)	6.6	5.5	7.0	3.9		2.7		9.0	5.5	2.0	4.0	20.0	5.0	7.6	3.0	3.0	3.3
Average nominal interest rate on forex debt (in percent)	2.7	2.6	2.5	3.1		0.4		2.7	2.8	2.8	2.9	2.9	2.9	2.8	2.7	2.5	2.6
Average real interest rate on domestic debt (in percent)	0.9	12.8	0.3	3.9		5.9		-4.1	-0.2	3.0	1.5	-12.3	3.6	-1.4	5.3	5.3	4.5
Real exchange rate depreciation (in percent, + indicates depreciation)	-11.4	5.3	-6.8	-5.0		10.2		0.0	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	7.7	-2.5	7.0	6.0		4.2		12.5	9.6	5.5	6.9	24.6	7.5	11.1	4.0	4.0	4.8
Growth of real primary spending (deflated by GDP deflator, in percent)	0.0	0.5	0.0	0.1		0.2		0.1	0.0	0.0	0.0	-0.1	-0.1	0.0	0.0	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	...	…		…		32.0	30.2	28.0	26.4	25.3	22.7	27.4	22.4	18.6	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Appendix Table 2 Papua New Guinea: Sensitivity Analysis for Key Indicators of Public Debt, 2011–2031

	Projections							
	2011	2012	2013	2014	2015	2016	2021	2031
PV of Debt-to-GDP Ratio								
Baseline	22	18	18	19	14	13	10	7
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	22	17	13	8	-3	-8	-32	-66
A2. Primary balance is unchanged from 2011	22	18	16	12	2	-1	-17	-36
A3. Permanently lower GDP growth 1/	22	19	19	20	15	15	16	31
A4. All Other Liabilities (Pensions, SOEs and LNG Contingent Liabilities).	22	26	26	27	26	26	22	17
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012-2013	22	20	22	24	19	19	20	30
B2. Primary balance is at historical average minus one standard deviations in 2012-2013	22	22	24	26	20	19	16	14
B3. Combination of B1-B2 using one half standard deviation shocks	22	20	19	21	16	15	15	18
B4. One-time 30 percent real depreciation in 2012	22	23	22	23	18	17	14	13
B5. 10 percent of GDP increase in other debt-creating flows in 2012	22	29	29	30	23	23	20	19
PV of Debt-to-Revenue Ratio 2/								
Baseline	66	58	61	77	79	78	55	44
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	66	53	44	31	-14	-46	-172	-395
A2. Primary balance is unchanged from 2011	66	56	53	48	13	-8	-93	-218
A3. Permanently lower GDP growth 1/	66	59	63	81	87	88	85	186
A4. All Other Liabilities (Pensions, SOEs and LNG Contingent Liabilities).	66	82	88	107	147	155	119	105
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012-2013	66	64	73	95	107	111	110	179
B2. Primary balance is at historical average minus one standard deviations in 2012-2013	66	70	82	102	111	111	87	87
B3. Combination of B1-B2 using one half standard deviation shocks	66	62	64	83	90	91	79	108
B4. One-time 30 percent real depreciation in 2012	66	72	75	93	100	99	75	77
B5. 10 percent of GDP increase in other debt-creating flows in 2012	66	91	97	120	133	134	109	117
Debt Service-to-Revenue Ratio 2/								
Baseline	8	8	8	8	9	10	6	5
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	8	8	7	5	-6	-9	-49	-96
A2. Primary balance is unchanged from 2011	8	8	7	7	1	-2	-30	-55
A3. Permanently lower GDP growth 1/	8	8	8	9	11	11	11	36
A4. All Other Liabilities (Pensions, SOEs and LNG Contingent Liabilities).	8	8	11	16	18	31	21	17
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012-2013	8	8	8	10	15	16	17	36
B2. Primary balance is at historical average minus one standard deviations in 2012-2013	8	8	9	13	20	18	13	16
B3. Combination of B1-B2 using one half standard deviation shocks	8	8	8	9	11	12	11	20
B4. One-time 30 percent real depreciation in 2012	8	9	9	10	13	13	10	14
B5. 10 percent of GDP increase in other debt-creating flows in 2012	8	8	11	19	27	25	19	23

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.

Appendix Table 3a — Papua New Guinea: External Debt Sustainability Framework, Baseline Scenario, 2008–2031 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average	6/	Standard Deviation	6/	Projections						2011-2016			2017-2031
	2008	2009	2010					2011	2012	2013	2014	2015	2016	Average	2021	2031	Average
External debt (nominal) 1/	**25.6**	**26.6**	**23.9**					**22.8**	**23.8**	**25.6**	**26.5**	**21.0**	**21.2**		**24.3**	**28.4**	
o/w public and publicly guaranteed (PPG)	13.2	12.9	10.7					9.7	9.4	9.7	9.7	7.2	6.9		6.4	5.6	
Change in external debt	-5.7	1.1	-2.7					-1.1	1.0	1.8	0.9	-5.4	0.2		1.0	0.4	
Identified net debt-creating flows	-16.0	2.8	0.9					2.7	2.2	1.6	0.6	-13.0	-13.5		-11.8	-7.6	
Non-interest current account deficit	**-11.5**	**6.6**	**22.3**	**-2.9**		**10.4**		**19.9**	**13.8**	**7.1**	**4.3**	**-8.2**	**-12.7**		**-12.1**	**-8.7**	-11.7
Deficit in balance of goods and services	-16.1	1.7	9.7					-0.6	-5.4	-8.2	-8.1	-21.5	-28.1		-24.7	-17.4	
Exports	77.3	56.8	63.6					75.1	73.7	68.0	61.7	58.3	60.9		49.5	39.3	
Imports	61.2	58.4	73.2					74.4	68.3	59.8	53.6	36.8	32.8		24.8	21.9	
Net current transfers (negative = inflow)	-2.0	-1.4	-2.9	-4.6		2.3		-2.5	-1.9	-1.5	-1.2	-0.8	-0.6		-0.2	0.2	-0.1
o/w official	-1.6	-4.0	-5.5					-5.0	-4.4	-4.1	-3.9	-2.7	-2.4		-1.8	-1.1	
Other current account flows (negative = net inflow)	6.7	6.3	15.6					23.1	21.1	16.8	13.7	14.1	15.9		12.8	8.5	
Net FDI (negative = inflow)	**0.4**	**-5.3**	**-19.0**	**-4.7**		**5.5**		**-17.0**	**-12.1**	**-6.9**	**-4.7**	**-2.7**	**-1.6**		**-1.2**	**-0.8**	-1.1
Endogenous debt dynamics 2/	**-4.9**	**1.5**	**-2.5**					**-0.2**	**0.6**	**1.4**	**1.0**	**-2.1**	**0.8**		**1.5**	**1.9**	
Contribution from nominal interest rate	1.6	1.7	1.6					1.5	1.7	1.9	2.0	1.6	1.7		2.1	2.7	
Contribution from real GDP growth	-1.6	-1.4	-1.6					-1.8	-1.2	-0.5	-1.0	-3.7	-1.0		-0.7	-0.8	
Contribution from price and exchange rate changes	-4.8	1.2	-2.5					…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**10.3**	**-1.8**	**-3.6**					**-3.8**	**-1.2**	**0.1**	**0.3**	**7.5**	**13.7**		**12.8**	**8.0**	
o/w exceptional financing	0.0	0.0	0.0					0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	24.4					23.0	23.8	25.3	26.0	20.5	20.7		23.5	27.6	
In percent of exports	...	...	38.3					30.7	32.3	37.2	42.1	35.2	34.0		47.4	70.3	
PV of PPG external debt	**...**	**...**	**11.1**					**9.8**	**9.3**	**9.4**	**9.2**	**6.7**	**6.4**		**5.7**	**4.8**	
In percent of exports	**...**	**...**	**17.5**					**13.1**	**12.6**	**13.8**	**14.9**	**11.4**	**10.5**		**11.4**	**12.1**	
In percent of government revenues	**...**	**...**	**41.2**					**34.7**	**34.4**	**36.9**	**43.0**	**44.8**	**43.9**		**33.9**	**30.9**	
Debt service-to-exports ratio (in percent)	**5.1**	**7.2**	**5.5**					**4.3**	**4.2**	**5.3**	**6.0**	**4.9**	**4.8**		**6.5**	**9.5**	
PPG debt service-to-exports ratio (in percent)	**2.0**	**2.6**	**1.8**					**1.3**	**1.2**	**1.3**	**1.4**	**1.0**	**0.9**		**0.8**	**1.0**	
PPG debt service-to-revenue ratio (in percent)	**5.7**	**6.4**	**4.1**					**3.5**	**3.3**	**3.5**	**4.0**	**4.1**	**3.9**		**2.4**	**2.5**	
Total gross financing need (Billions of U.S. dollars)	-0.3	0.7	1.0					1.1	1.0	1.0	1.0	-0.9	-1.4		-1.2	0.8	
Non-interest current account deficit that stabilizes debt ratio	-5.8	5.6	25.0					21.0	12.8	5.3	3.4	-2.8	-12.9		-13.1	-9.1	
Key macroeconomic assumptions																	
Real GDP growth (in percent)	6.6	5.5	7.0	3.9		2.7		9.0	5.5	2.0	4.0	20.0	5.0	7.6	3.0	3.0	3.3
GDP deflator in US dollar terms (change in percent)	18.3	-4.4	10.4	6.7		10.0		9.9	2.5	-0.3	2.1	19.4	2.6	6.1	0.4	1.9	2.1
Effective interest rate (percent) 5/	6.4	6.6	7.0	5.8		1.7		7.7	8.2	8.1	8.2	8.8	8.9	8.3	9.5	10.2	9.7
Growth of exports of G&S (US dollar terms, in percent)	20.6	-26.0	32.2	11.5		22.1		41.5	6.1	-6.0	-3.7	35.4	12.5	14.3	0.4	2.4	2.4
Growth of imports of G&S (US dollar terms, in percent)	4.9	-3.7	48.0	14.0		19.5		21.7	-0.8	-10.9	-4.8	-1.7	-3.8	-0.1	3.0	3.8	2.7
Grant element of new public sector borrowing (in percent)	...	...	...	...		...		32.0	30.2	28.0	26.4	25.3	22.7	27.4	22.4	18.6	21.2
Government revenues (excluding grants, in percent of GDP)	27.2	23.5	27.0					28.4	27.0	25.4	21.3	14.9	14.5		16.7	15.4	15.7
Aid flows (in Billions of US dollars) 7/	0.4	0.4	0.6					0.6	0.6	0.6	0.6	0.6	0.5		0.6	0.6	
o/w Grants	0.4	0.3	0.5					0.6	0.5	0.5	0.5	0.5	0.5		0.5	0.5	
o/w Concessional loans	0.0	0.1	0.1					0.1	0.1	0.1	0.1	0.1	0.0		0.1	0.1	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...					5.3	4.8	4.5	4.2	2.9	2.5		1.9	1.2	1.7
Grant-equivalent financing (in percent of external financing) 8/	...	...	...					88.3	86.2	84.5	82.7	82.5	83.9		81.6	78.9	80.8
Memorandum items:																	
Nominal GDP (Billions of US dollars)	8.0	8.1	9.5					11.4	12.3	12.5	13.3	19.1	20.6		28.1	45.6	
Nominal dollar GDP growth	26.2	0.8	18.1					19.8	8.1	1.7	6.2	43.3	7.7	14.5	3.4	5.0	5.5
PV of PPG external debt (in Billions of US dollars)			1.1					1.1	1.1	1.1	1.2	1.2	1.3		1.6	2.2	
(PVt-PVt-1)/GDPt-1 (in percent)								0.1	0.2	0.3	0.4	0.4	0.2	0.3	0.2	0.1	0.2
Gross workers' remittances (Billions of US dollars)	0.0	0.0	0.0					0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of PPG external debt (in percent of GDP + remittances)	...	...	11.1					9.8	9.3	9.4	9.2	6.7	6.4		5.7	4.8	
PV of PPG external debt (in percent of exports + remittances)	...	...	17.5					13.1	12.6	13.8	14.9	11.4	10.5		11.4	12.1	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	1.8					1.3	1.2	1.3	1.4	1.0	0.9		0.8	1.0	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Appendix Table 3b Papua New Guinea: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2011–2031

(In percent)

	Projections							
	2011	2012	2013	2014	2015	2016	**2021**	2031
PV of debt-to GDP ratio								
Baseline	10	9	9	9	7	6	**6**	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	10	0	-8	-16	-14	-10	**13**	15
A2. New public sector loans on less favorable terms in 2011-2031 2	10	9	10	10	7	7	**7**	8
A3. Authorities' Price Assumptions for Gold, Copper and Oil in 2011-2013.	9	22	38	37	26	25	**18**	5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	10	9	9	9	6	5	**3**	-2
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	10	19	31	30	21	20	**14**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	10	9	9	9	6	6	**3**	-2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	10	17	21	20	14	13	**9**	1
B5. Combination of B1-B4 using one-half standard deviation shocks	10	21	24	23	17	16	**11**	2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	10	12	12	11	8	7	**4**	-3
PV of debt-to-exports ratio								
Baseline	13	13	14	15	11	10	**11**	12
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	13	0	-12	-25	-24	-16	**27**	39
A2. New public sector loans on less favorable terms in 2011-2031 2	13	13	14	16	13	12	**15**	19
A3. Authorities' Price Assumptions for Gold, Copper and Oil in 2011-2013.	13	39	78	60	45	41	**36**	13
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	13	12	13	13	10	8	**5**	-5
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	13	31	58	61	46	42	**37**	11
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	13	12	13	13	10	8	**5**	-5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	13	23	31	33	25	22	**19**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	13	29	35	37	28	25	**21**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	13	12	13	13	10	8	**5**	-5
PV of debt-to-revenue ratio								
Baseline	35	34	37	43	45	44	**34**	31
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	35	-1	-33	-73	-93	-67	**80**	100
A2. New public sector loans on less favorable terms in 2011-2031 2	35	34	38	46	49	50	**43**	49
A3. Authorities' Price Assumptions for Gold, Copper and Oil in 2011-2013.	33	82	150	172	176	171	**108**	34
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	35	34	36	41	41	37	**17**	-14
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	35	70	121	139	142	138	**85**	22
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	35	35	37	42	42	38	**18**	-14
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	35	64	83	95	97	93	**55**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	35	77	96	110	112	108	**64**	10
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	35	46	48	54	53	49	**23**	-18
Debt service-to-exports ratio								
Baseline	1	1	1	1	1	1	**1**	1
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	1	1	1	0	0	0	**0**	2
A2. New public sector loans on less favorable terms in 2011-2031 2	1	1	1	1	1	1	**1**	1
A3. Authorities' Price Assumptions for Gold, Copper and Oil in 2011-2013.	1	2	3	3	2	2	**3**	2
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	1	1	1	1	1	1	**1**	0
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	1	1	2	3	2	2	**3**	2
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	1	1	1	1	1	1	**1**	0
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	1	1	2	2	1	1	**2**	1
B5. Combination of B1-B4 using one-half standard deviation shocks	1	1	2	2	2	1	**2**	1
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	1	1	1	1	1	1	**1**	0
Debt service-to-revenue ratio								
Baseline	4	3	4	4	4	4	**2**	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	4	3	2	1	0	1	**-1**	6
A2. New public sector loans on less favorable terms in 2011-2031 2	4	3	3	4	4	4	**3**	4
A3. Authorities' Price Assumptions for Gold, Copper and Oil in 2011-2013.	4	3	5	8	8	8	**9**	5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	4	4	4	4	4	4	**2**	0
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	4	3	5	7	7	7	**7**	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	4	4	4	4	4	4	**2**	0
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	4	3	4	6	6	5	**5**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	4	4	5	6	6	6	**6**	3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	4	5	5	5	6	5	**3**	1
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	19	19	19	19	19	19	**19**	19

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

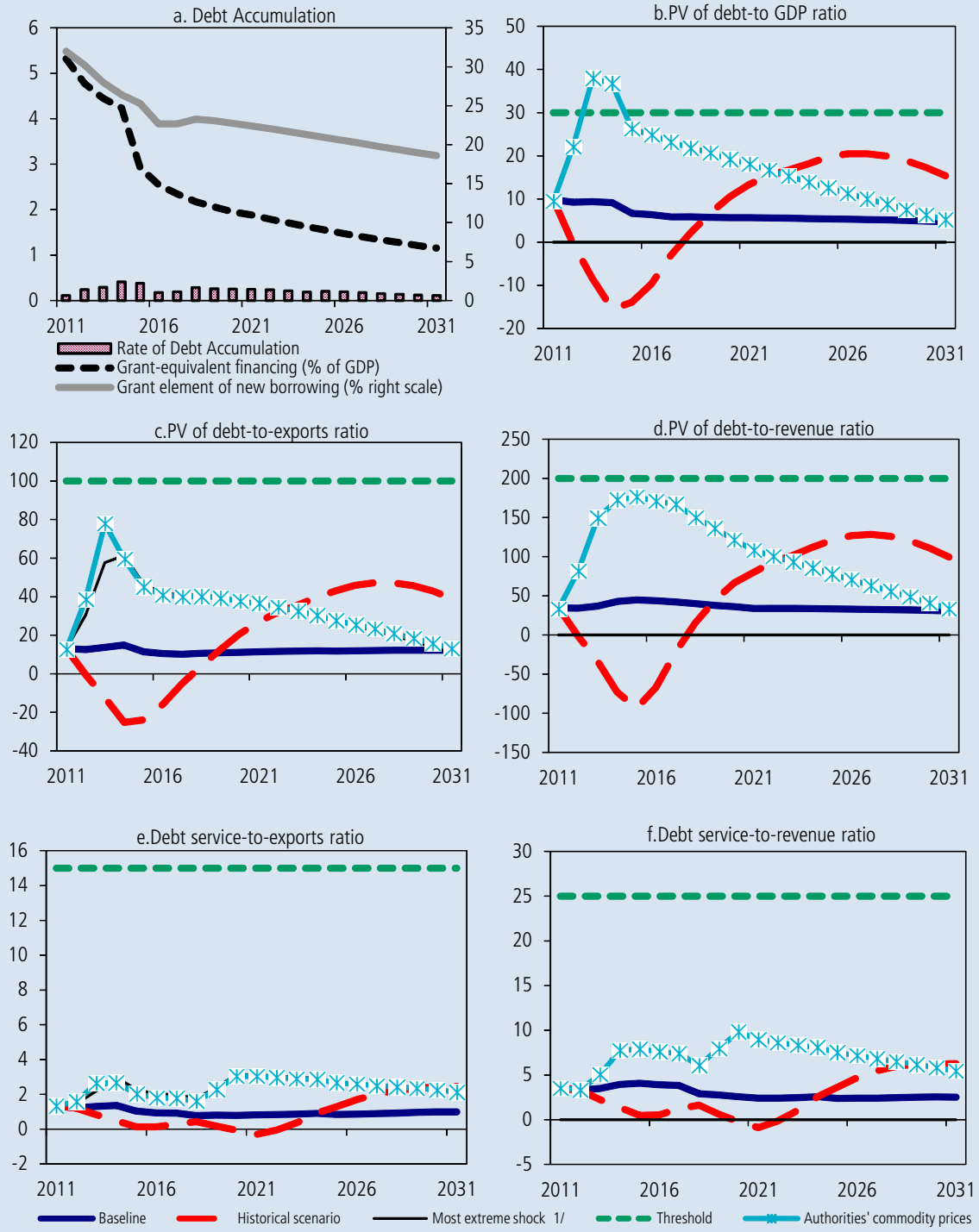
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

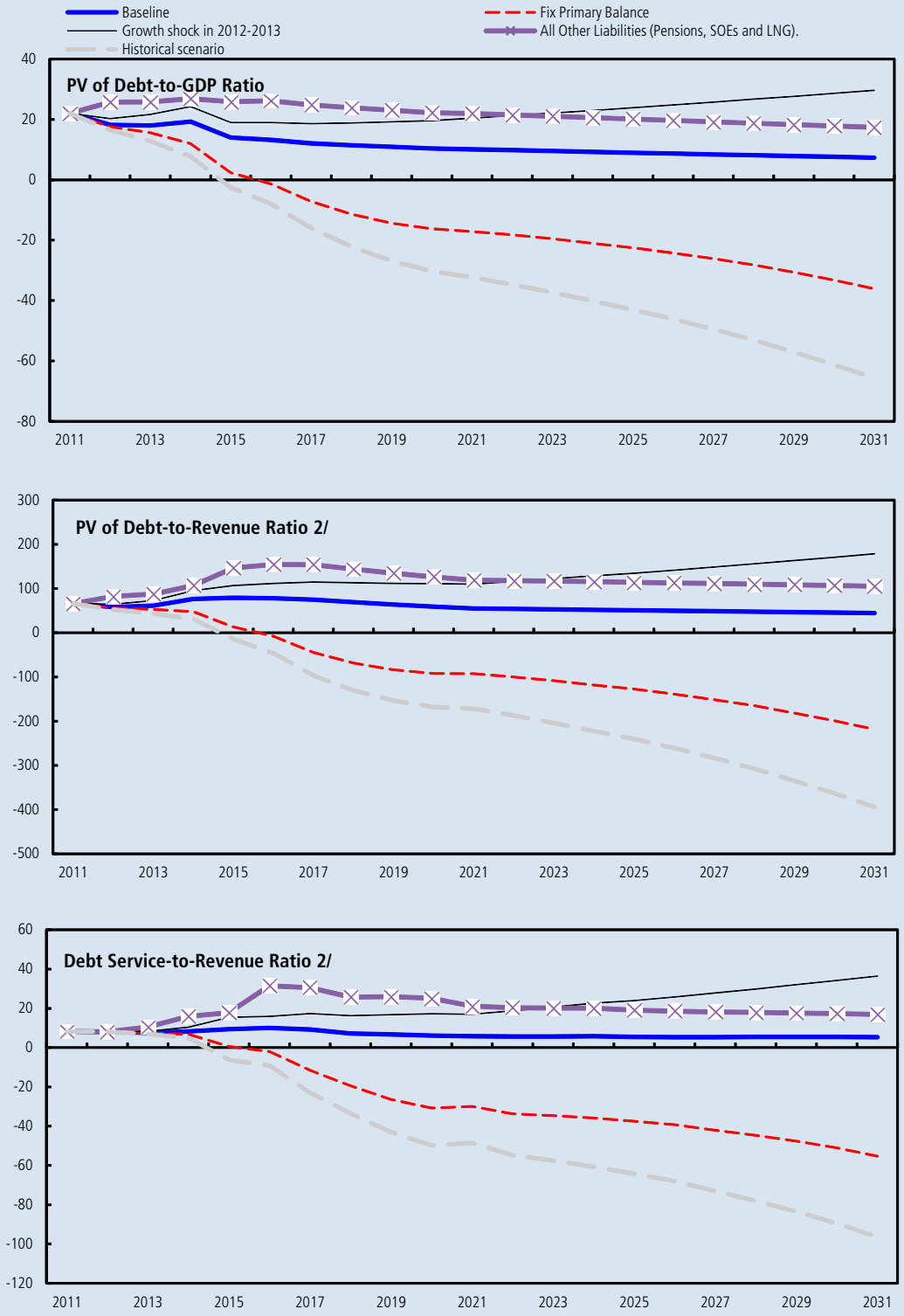
Appendix Figure 1 Papua New Guinea: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2011–2031 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2021. In figure b. it corresponds to an authorities' prices shock; in c. to a Exports shock; in d. to an authorities' prices shock; in e. to a Exports shock and in figure f. to an authorities' prices shock

Appendix Figure 2 Papua New Guinea: Indicators of Public Debt Under Alternative Scenarios, 2011–2031 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2021.
2/ Revenues are defined inclusive of grants.

Appendix 4 Papua New Guinea—FSAP: Main Lessons and Take-Aways

The joint IMF-World Bank FSAP mission, which visited Port Moresby in May 2010, took place against (i) a backdrop of wide-spread optimism for the operation of the Liquefied Natural Gas (LNG) projects after 2014 and (ii) a mild impact of the financial crisis on PNG largely due to the broad insulation of the financial sector from the global financial markets. The main findings and recommendations, which were updated during the 2011 Article IV consultations, are summarized below.

PNG's financial sector

Total financial sector assets have grown to roughly 95 percent of GDP and are held by a wide range of institutions. Currently, there are four commercial banks, seven authorized superannuation funds, and a number of other nonbank financial sector institutions licensed and regulated by BPNG. General insurance companies regulated by the Insurance Commission and the stock exchange by the Securities Commission.

Banks and superannuation funds dominate the financial sector. Since the financial sector reforms in early 2000, banks have increased their assets fivefold to almost K17 billion by end-2010. The nationally owned Bank of South Pacific is the largest, followed by subsidiaries of two Australian banks and a Malaysian bank. In addition, there is a state-owned, non-deposit taking National Development Bank, which has a primary function of providing accessible credit to people in rural areas. Nonbanks are important, too, with about K7.5 billion in assets. Among them, the authorized superannuation funds (ASFs) are the largest, holding more than 70 percent of total nonbank financial assets. ASFs, in terms of assets size, are followed by the savings and loans societies, finance and microfinance companies, and insurance companies (including general, life, and broker).

Financial Sector Institutions	Total Assets	
	In million Kina	In percent
Commercial Banks	16715.3	68.3
National Development Bank	200.5	0.8
Authorized Superannuation Funds	5450.8	22.3
Other Nonbank Institutions	2097.1	7.9
Total	24463.7	100

Source: BPNG and IMF staff estimates as of September 2010.

Access to finance is one of the major hurdles to further financial sector deepening. Only about 10 percent of the population and 17 percent of adults currently have access to formal financial services. Although financial services providers make efforts to expand their services, the majority of people in rural communities are still financially excluded. The staff supports the authorities' plan to conduct a financial literacy survey in partnership with the World Bank and UNDP during 2011. This survey could lead to specific projects aimed at extending financial services to complement the banking services provided by microfinance institutions to people in rural areas.

The authorities are committed to support financial inclusion. The PNG Development Strategic Plan 2010–2030 identifies the extension of microfinance banking services to all districts in PNG as a priority. In addition, the national payment system development program aims to encourage the introduction and use of innovative, convenient, and cost-effective new retail payment instruments, with a special focus on rural areas.

Overall financial sector soundness is satisfactory

Indicators point to a generally well-capitalized and highly profitable financial sector. Credit and market risks in the banking sector are relatively low and can easily be offset by prevailing high levels of capital, low levels of nonperforming loans, and high provisions. Nonetheless, high capital ratios are appropriate to accommodate lending to larger clients and growing credit risk that is expected in the medium term. Also, high concentration of the financial sector, the absence of secondary market for government securities, and the lack of repo and discount windows require banks to maintain higher liquidity.

Regulation and supervision of the banking sector has been strengthened, although several elements will still be required to move toward a more risk-based approach. The BPNG has been developing its supervisory techniques since the reforms of 2000, but it is essential that it completes the suite of prudential standards, enhance the available supervisory tools, and make better use of the tools it has. The crisis management framework is incomplete, and steps need to be taken to put this in place.

Authorized Superannuation Funds (ASFs) are the principal source of long-term investment. Papua New Guinea's demographic profile and a system of compulsory savings in the formal sector, along with wage growth, will ensure a positive net cash flow and

Appendix 4 Papua New Guinea—FSAP: Main Lessons and Take-Aways

(continued)

anchor the role of these funds as a key source of domestic savings. The ASFs and general insurers have been performing with increasing efficiency, while general insurers are well capitalized, profitable and exhibit sound risk management.

The reform agenda needs to continue

Payment and settlements systems need major reforms. The legal framework does not recognize payment services as an independent activity, while the lack of an inter-bank real-time payment system with collateralized liquidity management creates systemic risks. The clearing house operations are manual and paper based. Establishing an independent Payment Systems Department in the BPNG is crucial to support regulation and oversight.

Efforts to develop the local government securities markets need to be intensified. Although the overall operation of the primary market reflects best practice, banks have limited incentives to canvass investors. Smaller investors face hurdles, as the minimum amount to participate on a competitive basis is high and there is no opportunity for smaller investors to enter noncompetitive bids. Exposure of PNG to international investors through the LNG projects may provide new opportunities to attract nonresident investors.

Improving access to financial services remains a developmental challenge, and efforts need to be stepped up. Around 85 percent of the adult population is excluded from the formal sector, largely those in rural communities. The main barriers are operational challenges and infrastructure weaknesses, resulting in high intermediation costs.

Further developments on AML and CFT issues should be pursued. This should include enacting legislation on terrorist financing and the use of existing supervisory powers to encourage banks to undertake detailed customer due diligence.

The establishment of a sovereign wealth fund is under consideration, but modalities need to be carefully designed. In view of the large revenues anticipated from the LNG projects, a key concern for the government is the setting-up of the appropriate institutional and investment management arrangements.

Attachment	Main FSAP Recommendations	
		Priority
Banking Sector		
Credit Risk		
•	Monitor current and emerging NPLs to assess the causes and macroeconomic impacts.	High
•	Conduct a thorough analysis of counterparty exposure concentrations, with full account of interconnectedness.	Medium
•	Conduct a thorough analysis of collateral and related exposures of lending to property valuations.	Medium
Liquidity Risk		
•	Develop monitoring systems and stress testing that allow for a revised definition of liquid assets (without government securities).	High
•	Ensure that any steps to remove government deposits from banks are done with careful planning and consultation.	High
•	Clarify and formalize bank liquidity-support arrangements for each bank with the BPNG, including through repurchase arrangements for government securities.	High
Regulation and Supervision		
•	Complete the supervisory regime by publishing a full set of prudential standards, starting with risk management, market, and liquidity risk but also including governance, credit, and operational risk in the context of a phased move to Basel II.	High
•	Increase the range of administrative sanctions and make compliance with prudential standards compulsory, while developing a remedial action program.	High
•	Enhance the capacity of the supervisory staff through training, so that the BPNG can move to full risk-based supervision.	High
•	Conduct and document a full risk assessment as the basis for supervisory strategy.	Medium
•	BPNG to explore scope for technical assistance to centralize data management systems.	Medium
Systemically important institutions		
•	Arrangements for liquidity improvement and contingency planning should be made to reduce the risk profile, while the number of shareholders should be increased in a measured way.	Medium
Sovereign Wealth Funds		
•	If authorities opt for a SWF to manage future LNG project-related proceeds, decide on the objective(s), funding and withdrawal rules, institutional arrangements, including organizational structure, roles and responsibilities, and investment strategy.	High
Government Debt Markets		
•	Assess the broader public-sector debt obligations such as borrowing and guarantees.	High
•	Issue BPNG guidelines to market participants on sound procedures for repurchase transactions.	High
•	Introduce a requirement that all repos should be collateralized and introduce the Master Repurchase Agreement.	High
•	Transfer trust funds accounts to BPNG whilst carefully considering bank positions.	Medium

Attachment Main FSAP Recommendations

	Priority
Payment Systems	
• Implement the proposed National Payments System Development Plan.	High
• Establish a Payment Systems Department in the BPNG, with oversight of all payment and securities settlement systems.	High
Insurance Sector	
• Enhance co-operation with local and foreign supervisory authorities, and improve supervisory transparency.	High
• Enhance the supervisory functions regarding offsite reporting and monitoring and onsite inspections.	Medium
• Develop guidance on governance, risk management, and internal controls.	Medium
Financial Inclusion	
• Establish a functional coordination mechanism on financial inclusion through a national consultative process.	Medium
AML	
• Give BPNG responsibility for enforcing financial institutions' obligations.	High
• Complete and issue regulations on customer due diligence regime (especially with more focus on monitoring accounts).	High
Crisis Management	
• Establish a discount-window lending program, including repurchase agreements.	High
• Strengthen its crisis preparedness by developing a contingency planning framework, including internal procedures on emergency liquidity assistance.	High